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Course: Cybersecurity

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Title: Lab 5: Network Intrusion Detection with pfSense and Suricata

Exercise 1: Accessing the pfSense Web Administration Interface

Objective

The objective of this exercise was to access the pfSense web administration interface, authenticate as the administrator, and complete the initial configuration wizard. The web interface provides centralized management of firewall settings, network interfaces, services, and security features that are required for the remaining laboratory exercises.

Lab Execution Summary

After the pfSense installation was completed and the firewall rebooted successfully, the LAN interface IP address was displayed on the console. A client workstation connected to the LAN network was then used to access the pfSense web interface through a web browser. The administrator logged in using the default credentials and completed the initial setup wizard before reaching the pfSense Dashboard.

Implementation Steps

Step 1: Open the Web Browser

After the pfSense firewall finished booting, moved to the client workstation connected to the LAN network.

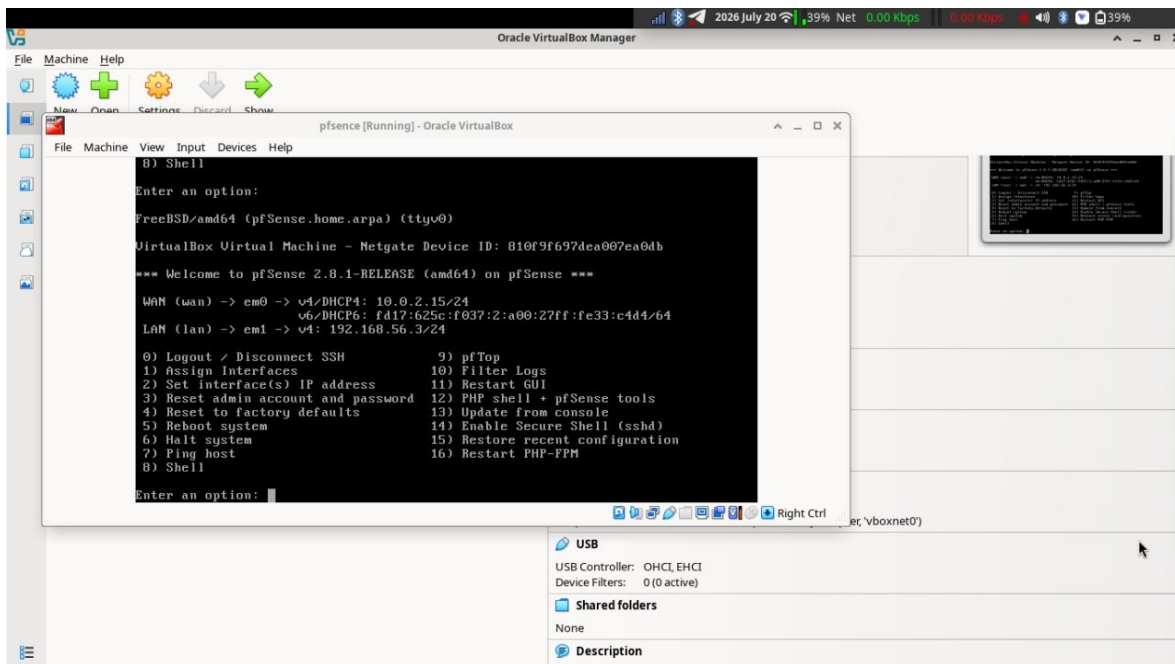
Opened a web browser (Google Chrome).

Step 2: Enter the pfSense Management IP Address

Typed the pfSense LAN management IP address into the browser's address bar.

Example:

<https://192.168.56.3>



Step 3: Accept the Security Certificate

Since pfSense uses a self-signed SSL certificate by default, the browser displayed a security warning indicating that the connection was not verified.

Selected:

- **Advanced**
- **Proceed to 192.168.1.1 (unsafe)** (Chrome)

to continue to the login page.

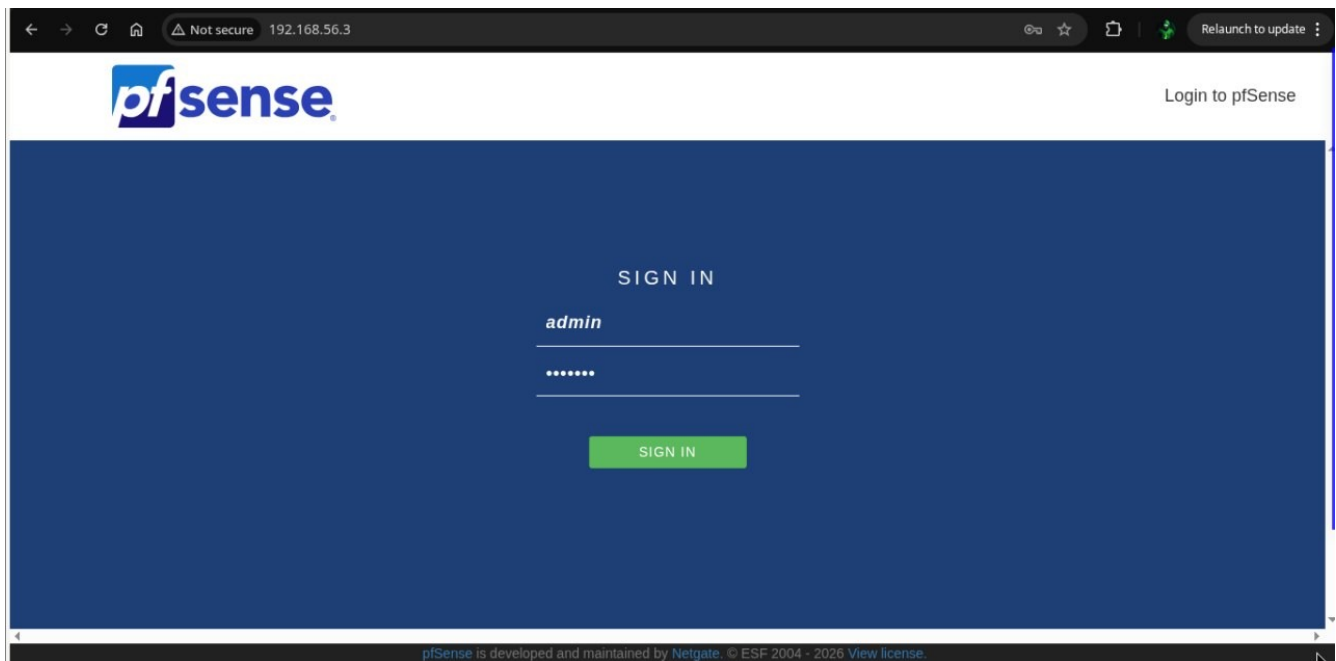
Step 4: Log in to pfSense

The pfSense login page appeared.

Entered the default administrator credentials.

Username Password

admin pfsense



Step 5: Start the Setup Wizard

After successful authentication, the **pfSense Setup Wizard** was displayed automatically.

Selected **Next** to begin the initial configuration process.

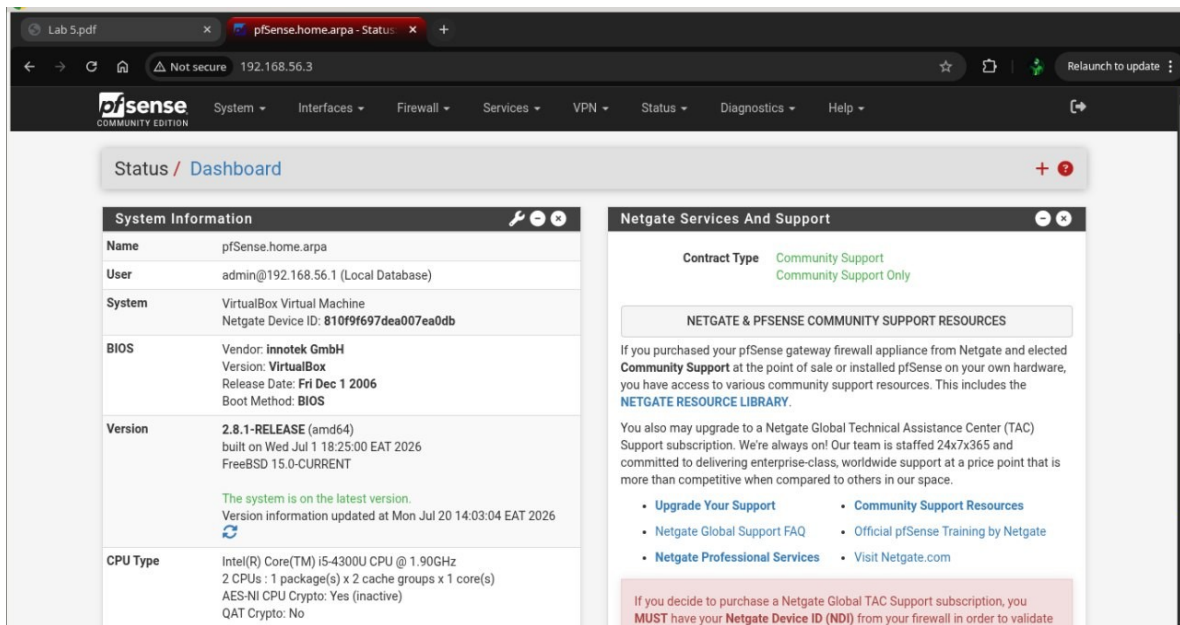
Step 6: Verify the pfSense Dashboard

After authentication, the pfSense Dashboard was displayed.

Verified the following information:

- System Information
- Firewall Status
- WAN Interface Status
- LAN Interface Status
- CPU Usage
- Memory Usage
- System Uptime

Confirmed that all interfaces were operating normally before proceeding.



Step 7: Verify the DNS Resolver

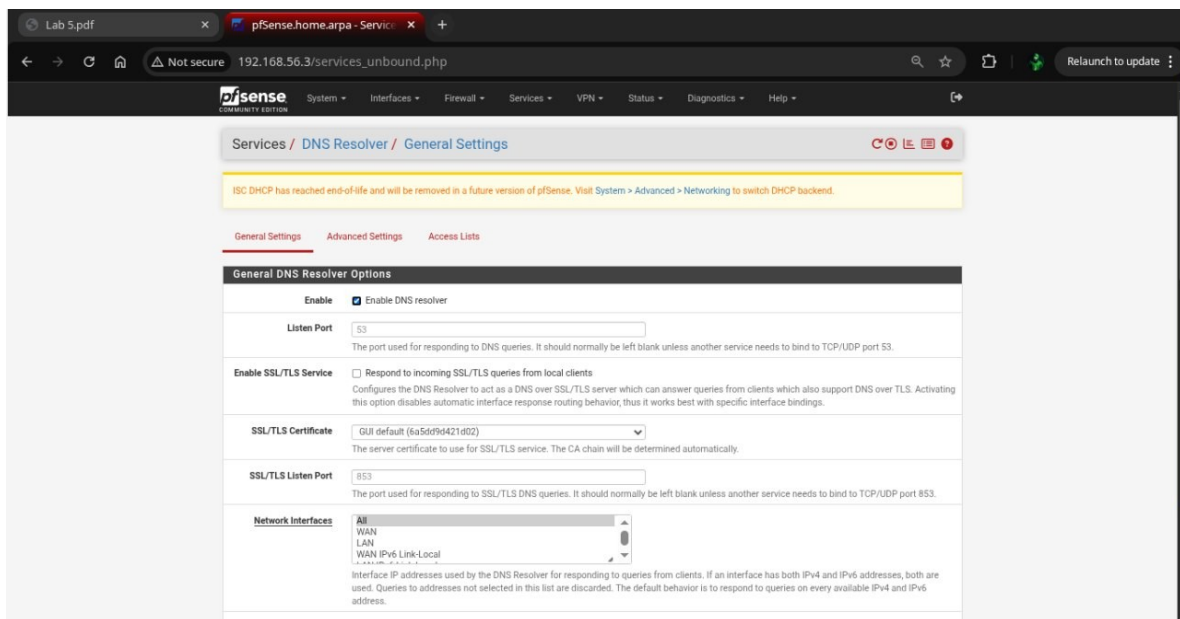
Navigated to:

Services → DNS Resolver

Verified that:

- DNS Resolver was enabled.
- Network interfaces were correctly selected.
- DNS resolution service was active.

This service allows pfSense to resolve domain names for clients connected to the network.



Step 8: Review General System Configuration

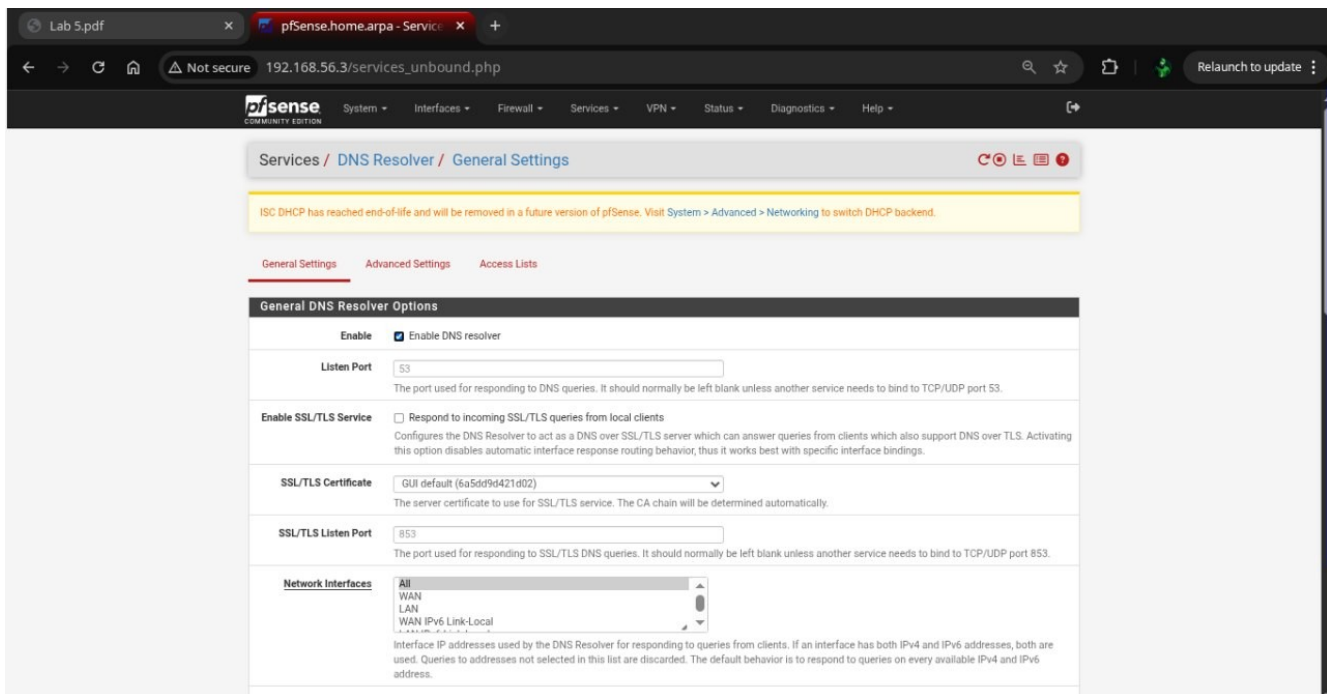
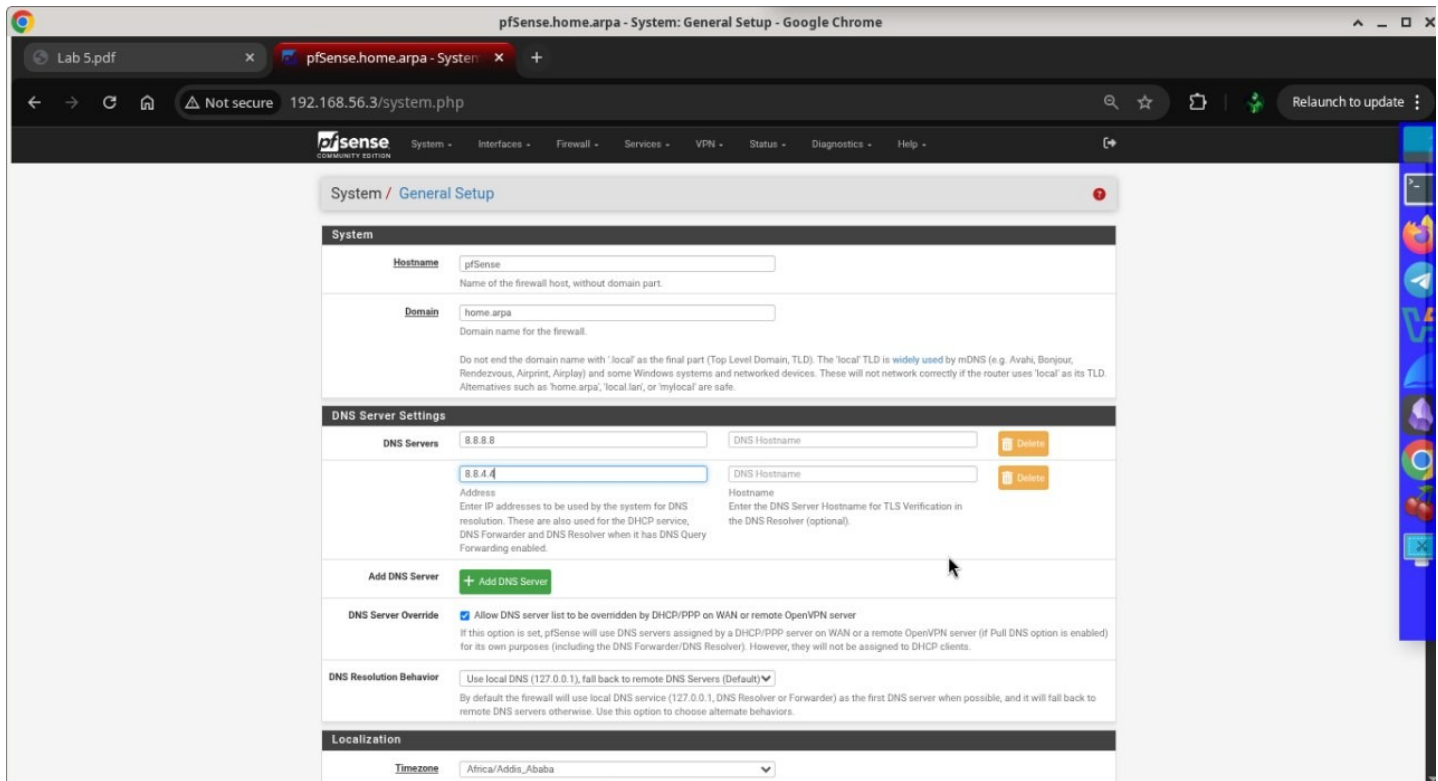
Navigated to:

System → **General Setup**

Verified the system configuration, including:

- Hostname
- Domain Name
- DNS Servers
- Time Zone

Confirmed that the configuration matched the laboratory environment.



pfSense.home.arpa - Services: DNS Resolver: General Settings - Google Chrome

Lab 5.pdf x pfSense.home.arpa - Services: x +

192.168.56.3/services_unbound.php

Relaunch to update

System Domain Local Zone Type

Transparent

The local-zone type used for the pfSense system domain ([System > General Setup](#)). Transparent is the default.

DNSSEC

☒ Enable DNSSEC Support

Python Module

☐ Enable Python Module

Enable the Python Module.

DNS Query Forwarding

☒ Enable Forwarding Mode

If this option is set, DNS queries will be forwarded to the upstream DNS servers defined under [System > General Setup](#) or those obtained via dynamic interfaces such as DHCP, PPP or OpenVPN (if DNS Server Override is enabled there).

☐ Use SSL/TLS for outgoing DNS Queries to Forwarding Servers

When set in conjunction with DNS Query Forwarding, queries to all upstream forwarding DNS servers will be sent using SSL/TLS on the default port of 853. Note that ALL configured forwarding servers MUST support SSL/TLS queries on port 853.

DHCP Registration

☐ Register DHCP leases in the DNS Resolver

If this option is set, then machines that specify their hostname when requesting an IPv4 DHCP lease will be registered in the DNS Resolver so that their name can be resolved. Note that this will cause the Resolver to reload and flush its resolution cache whenever a DHCP lease is issued. The domain in [System > General Setup](#) should also be set to the proper value.

Static DHCP

☐ Register DHCP static mappings in the DNS Resolver

If this option is set, then DHCP static mappings will be registered in the DNS Resolver, so that their name can be resolved. The domain in [System > General Setup](#) should also be set to the proper value.

OpenVPN Clients

☐ Register connected OpenVPN clients in the DNS Resolver

If this option is set, then the common name (CN) of connected OpenVPN clients will be registered in the DNS Resolver, so that their name can be resolved. This only works for OpenVPN servers (Remote Access SSL/TLS or User Auth with Username as Common Name option) operating in 'tun' mode. The domain in [System > General Setup](#) should also be set to the proper value.

Display Custom Options

[Display Custom Options](#)

Save

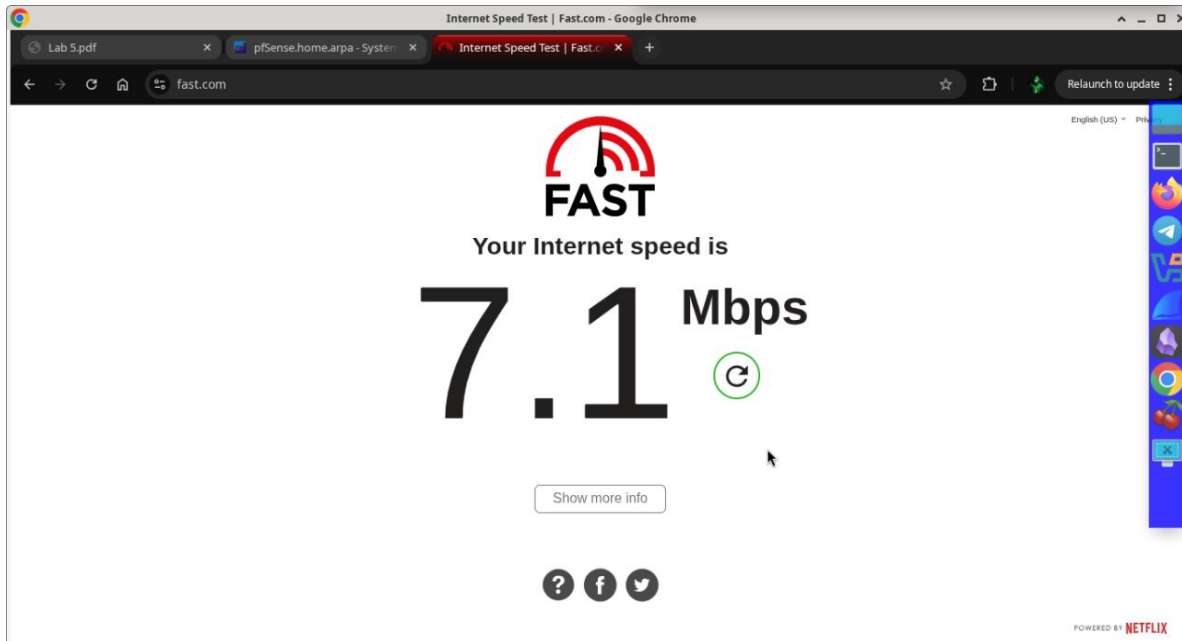
Host Overrides

Step 9: Test Internet Connectivity

Verified that the firewall had Internet connectivity.

Performed an Internet connectivity test (for example, using the Diagnostics menu or by opening an Internet speed test in the browser, depending on the lab procedure).

Confirmed that Internet access was available for downloading packages and updating security rules.



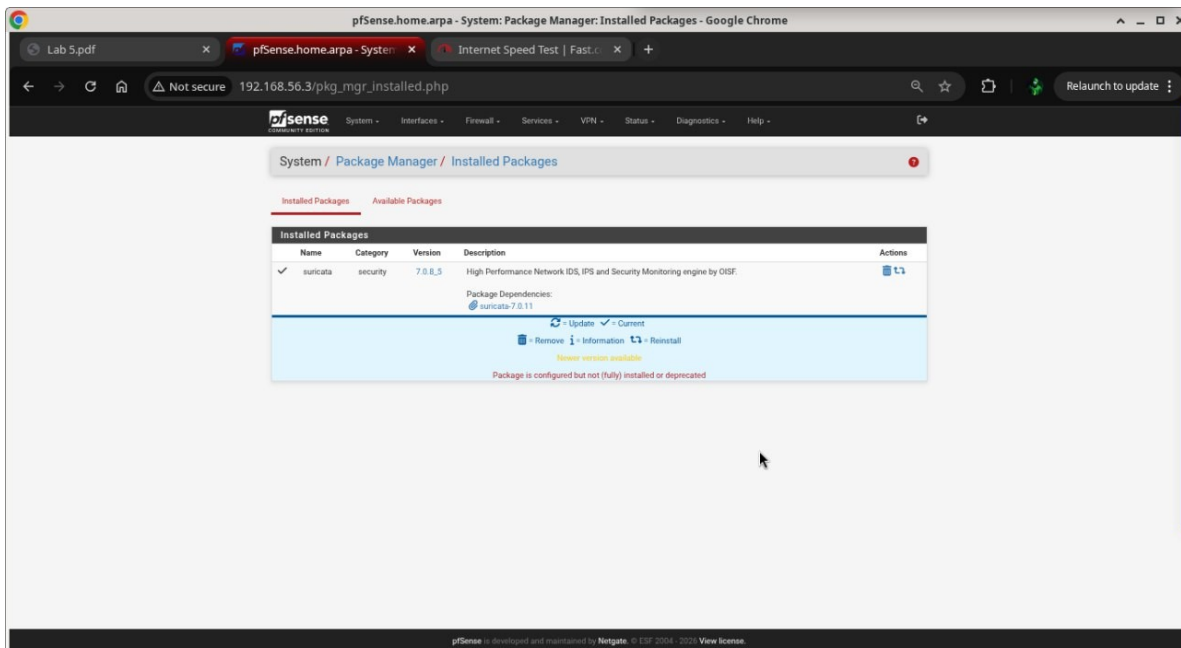
Step 10: Verify Installed Packages

Navigated to:

System → Package Manager → Installed Packages

Reviewed the list of installed packages.

Verified that the required packages for the laboratory were successfully installed.



Step 11: Open the Suricata Global Settings Page

Navigated to:

Services → Suricata → Global Settings

Reviewed the available global configuration options, including:

- Rule Update Settings
- Logging Options
- Automatic Updates
- General IDS Configuration

Verified that the Suricata package was correctly installed and accessible.

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pfSense.home.arpa - Services: Suricata: Global Settings

Internet Speed Test | Fast...

192.168.56.3/suricata/suricata_global.php

Relaunch to update

pfSense

Services / Suricata / Global Settings

InterfacesGlobal SettingsUpdatesAlertsBlocksFilesPass ListsSuppressLogs ViewLogs MgmtSID Mgmt

SyncIP Lists

Please Choose The Type Of Rules You Wish To Download

Install ETOpen Emerging Threats rules

☒ ETOpen is a free open source set of Suricata rules whose coverage is more limited than ETPro.

☐ Use a custom URL for ETOpen downloads

Enabling the custom URL option will force the use of a custom user-supplied URL when downloading ETOpen rules.

Install ETPro Emerging Threats rules

☐ ETPro for Suricata offers daily updates and extensive coverage of current malware threats.

☐ Use a custom URL for ETPro rule downloads

The ETPro rules contain all of the ETOpen rules, so the ETOpen rules are not required and are disabled when the ETPro rules are selected. [Sign Up for an ETPro Account.](#) Enabling the custom URL option will force the use of a custom user-supplied URL when downloading ETPro rules.

Install Snort rules

☐ Snort free Registered User or paid Subscriber rules

☐ Use a custom URL for Snort rule downloads

[Sign Up for a free Registered User Rules Account](#)
[Sign Up for paid Snort Subscriber Rule Set \(by Talos\)](#)

Enabling the custom URL option will force the use of a custom user-supplied URL when downloading Snort Subscriber rules.

Install Snort GPLv2 Community rules

☐ The Snort Community Ruleset is a GPLv2 Talos-certified ruleset that is distributed free of charge without any Snort Subscriber License restrictions.

☐ Use a custom URL for Snort GPLv2 rule downloads

This ruleset is updated daily and is a subset of the subscriber ruleset. If you are a Snort Subscriber Rules customer (paid subscriber), the community ruleset is already built into your download of the Snort Subscriber rules, and there is no benefit in adding this rule set separately.

Install Feodo Tracker Botnet C2 IP rules

☐ The Feodo Botnet C2 IP Ruleset contains Dridex and Emotet/Heodo botnet command and control servers (C&Cs) tracked by Feodo Tracker.

Install ABUSE.ch SSL Blacklist rules

☐ The ABUSE.ch SSL Blacklist Ruleset contains the SSL cert fingerprints of all SSL certs blacklisted by ABUSE.ch.

Hide Deprecated Rules Categories

☐ Hide deprecated rules categories in the GUI and remove them from the configuration. Default is Not Checked.

Download Extra Rules

☐ Download Extra Rules

Lab 5.pdf

pfSense.home.arpa - Service

Internet Speed Test | Fast.C

pfSense.home.arpa - Services: Suricata: Updates - Google Chrome

192.168.56.3/suricata/suricata_download_updates.php

Relaunch to update

pfSense

System

Interfaces

Firewall

Services

VPN

Status

Diagnostics

Help

Services / Suricata / Updates

Interfaces

Global Settings

Updates

Alerts

Sync

IP Lists

INSTALLED RULE SET MD5 SIGNATURES

Rule Set Name/Publisher	MD5 Signature Date
Emerging Threats Open Rules	Not Downloaded
Snort Subscriber Rules	Not Enabled
Snort GPLv2 Community Rules	Not Enabled
Feodo Tracker Botnet C2 IP Rules	Not Enabled
ABUSE.ch SSL Blacklist Rules	Not Enabled

UPDATE YOUR RULE SET

Last Update: Unknown

Result: Unknown

Update

Force

MANAGE RULE SET LOG

View Log

Log is empty.

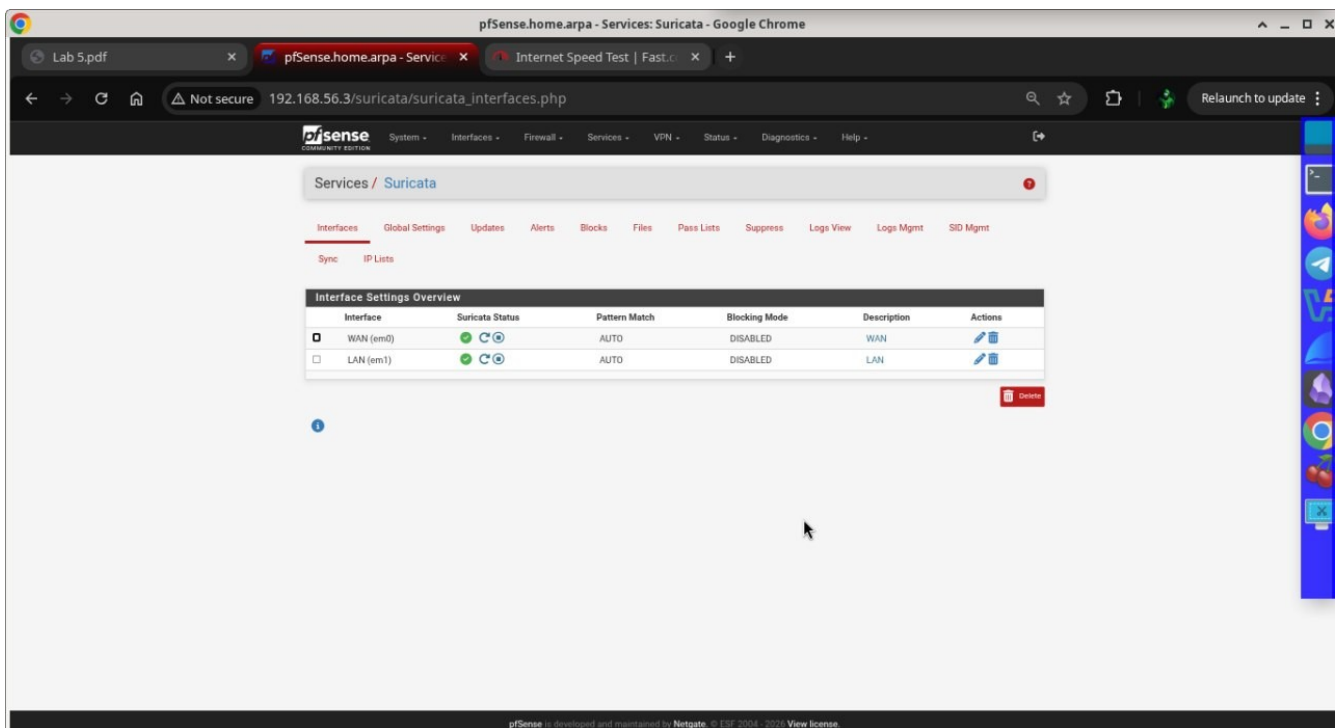
The log file is limited to 1024K in size and automatically clears when the limit is exceeded.

Rules Update Task

Updating rule sets may take a while - please wait for the process to complete.

This dialog will auto-close when the update is finished.

Close



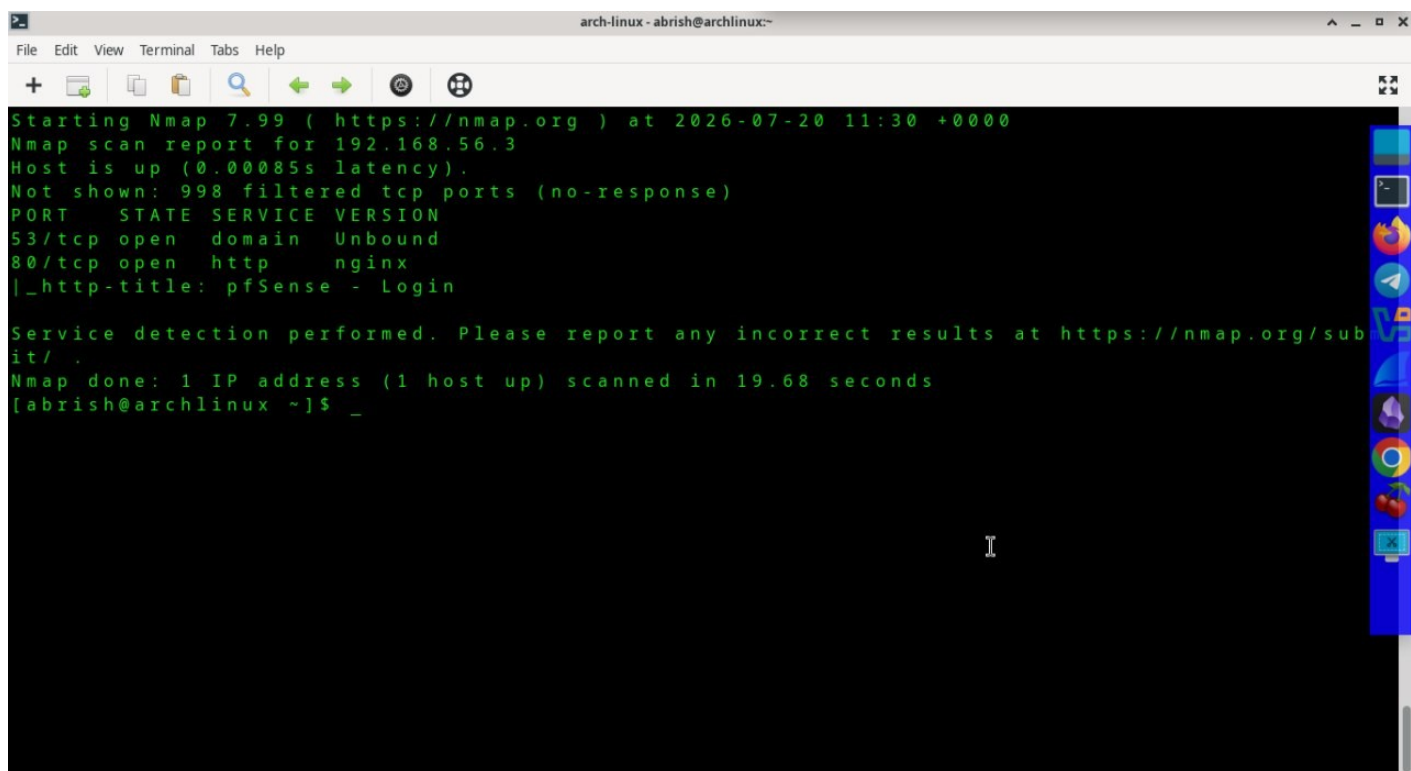
Step 12: Perform an Nmap Scan

From another machine on the network, performed an **Nmap** scan against the pfSense firewall or a monitored host to generate detectable network activity.

Example command:

```
nmap -sS 192.168.56.3
```

The scan generated network traffic that would later be analyzed by Suricata.



```
arch-linux - abrish@archlinux:~
File Edit View Terminal Tabs Help

Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 11:30 +0000
Nmap scan report for 192.168.56.3
Host is up (0.00085s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
53/tcp    open  domain  Unbound
80/tcp    open  http    nginx
|_http-title: pfSense - Login

Service detection performed. Please report any incorrect results at https://nmap.org/sub
it/ .
Nmap done: 1 IP address (1 host up) scanned in 19.68 seconds
[abrish@archlinux ~]$ _
```

Step 13: Review Suricata Alert Logs

Returned to the pfSense web interface.

Navigated to:

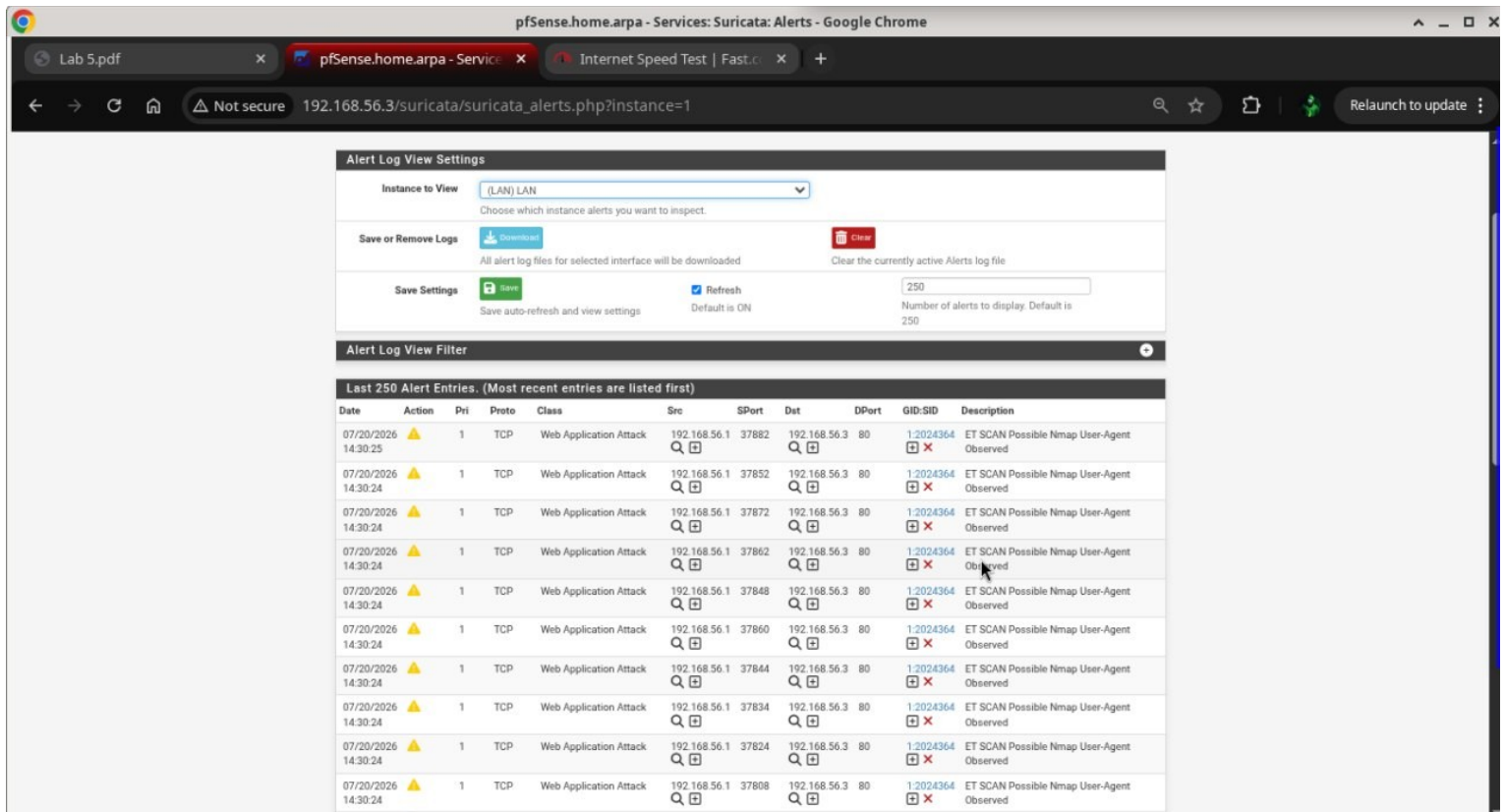
Services → Suricata → Alerts

Verified that the Nmap scan generated security alerts.

Reviewed the alert details, including:

- Timestamp
- Source IP Address
- Destination IP Address
- Alert Signature
- Priority
- Classification

Confirmed that Suricata successfully detected and logged the scanning activity.



The screenshot shows the pfSense web interface for Suricata Alerts. The page title is "pfSense.home.arpa - Services: Suricata: Alerts - Google Chrome". The browser address bar shows "192.168.56.3/suricata/suricata_alerts.php?instance=1". The page has a "Not secure" warning. The main content area is titled "Alert Log View Settings" and "Alert Log View Filter". Below these is a table of "Last 250 Alert Entries. (Most recent entries are listed first)".

Date	Action	Pri	Proto	Class	Src	SPort	Dest	DPort	UID:SID	Description
07/20/2026 14:30:25	⚠	1	TCP	Web Application Attack	192.168.56.1	37882	192.168.56.3	80	1:2024364	ET SCAN Possible Nmap User-Agent Observed
07/20/2026 14:30:24	⚠	1	TCP	Web Application Attack	192.168.56.1	37852	192.168.56.3	80	1:2024364	ET SCAN Possible Nmap User-Agent Observed
07/20/2026 14:30:24	⚠	1	TCP	Web Application Attack	192.168.56.1	37872	192.168.56.3	80	1:2024364	ET SCAN Possible Nmap User-Agent Observed
07/20/2026 14:30:24	⚠	1	TCP	Web Application Attack	192.168.56.1	37862	192.168.56.3	80	1:2024364	ET SCAN Possible Nmap User-Agent Observed
07/20/2026 14:30:24	⚠	1	TCP	Web Application Attack	192.168.56.1	37848	192.168.56.3	80	1:2024364	ET SCAN Possible Nmap User-Agent Observed
07/20/2026 14:30:24	⚠	1	TCP	Web Application Attack	192.168.56.1	37860	192.168.56.3	80	1:2024364	ET SCAN Possible Nmap User-Agent Observed
07/20/2026 14:30:24	⚠	1	TCP	Web Application Attack	192.168.56.1	37844	192.168.56.3	80	1:2024364	ET SCAN Possible Nmap User-Agent Observed
07/20/2026 14:30:24	⚠	1	TCP	Web Application Attack	192.168.56.1	37834	192.168.56.3	80	1:2024364	ET SCAN Possible Nmap User-Agent Observed
07/20/2026 14:30:24	⚠	1	TCP	Web Application Attack	192.168.56.1	37824	192.168.56.3	80	1:2024364	ET SCAN Possible Nmap User-Agent Observed
07/20/2026 14:30:24	⚠	1	TCP	Web Application Attack	192.168.56.1	37808	192.168.56.3	80	1:2024364	ET SCAN Possible Nmap User-Agent Observed

Conclusion

The initial pfSense configuration and system verification were completed successfully. The firewall web administration interface was accessed through a web browser, and the administrator authenticated successfully. The pfSense Dashboard was reviewed to confirm that the system and network interfaces were operating correctly. The DNS Resolver and General Setup configurations were verified, and Internet connectivity was confirmed to ensure the firewall could communicate with external networks and download required updates. The installed packages and Suricata Global Settings were also reviewed to verify that the intrusion detection system was available for configuration. Finally, an Nmap scan was performed to generate network traffic, and the resulting alerts were successfully detected and displayed in the Suricata Alerts log. These results confirmed that the pfSense firewall and Suricata IDS were functioning correctly and that the environment was fully prepared for subsequent intrusion detection and network security monitoring exercises.